A Novel EEG Signal Recognition Method Using Modified Optimal Electrodes Recombination Strategy

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Abstract. Electroencephalogram (EEG) is a comprehensive indicator of human physiological activities. Because of its comprehensiveness and complexities, an electrode-covered collection device on the scalp cannot collect the discharge phenomenon of the activated brain area exhaustively. A single electrode analysis will miss a lot of important association information between different brain regions. This paper presents a novel strategy to solve this problem by combining the optimal electrodes. The whole method is divided into five steps: (1) input multi-electrodes EEG data, (2) utilize the Principal Component Analysis (PCA) method to obtain the optimal electrodes, (3) adopt the modified optimal electrodes recombination strategy and obtain optimal electrode combination strategy, (4) extract features by using Empirical Mode Decomposition (EMD), and (5) use Naive Bayes classifier to do the classification tasks. In order to evaluate the validity of the proposed method, we apply the proposed method in BCI Competition II datasets Ia. Experimental results show that our method improves recognition performance for BCI Competition II datasets Ia and the modified optimal electrodes recombination strategy is reasonable. This provides a new idea for analyzing the EEG features of other tasks.

Keywords: Electroencephalogram, optimal electrodes selection, optimal electrodes recombination, Principal Component Analysis, Empirical Mode Decomposition.

1 Introduction

The brain is the controller of various human activities. When it produces mental activity, the cerebral cortex can produce a variety of waveforms; these EEG signals can be monitored externally. The study on brain development has been a continuous research work in recent decades. Brain Computer Interface (BCI) technologies make it possible to control the outside world through the advent of thought. It is a hot spot and

difficult point in the field of EEG study to improve classification precision of motor imagery BCI data.

An EEG signal is composed of a large number of brain cells signals superposition results. And thus, it is very complex because of its nonlinearity and overlap. This makes it difficult to obtain high signal accuracy in relation to the task being monitored since some signals are not of relevance.

The current EEG based BCI applications seek to recognize mental tasks by selecting electrodes related to these tasks, extracting relevant features from signals recorded through these electrodes, developing an algorithm that yields the highest classification rate, and providing this information to communication and control units through translation algorithms [1,2,3]. Most studies lay stress upon the feature extraction and classification algorithm. However, research shows that different mental tasks will activate special brain areas using functional magnetic resonance imaging (fMRI) [4] and Positron Emission Tomography (PET) technology [5]. What is more, the same mental task will activate different areas if experimental designs are different. Some experimental research demonstrates that the interaction effect exists between multi-areas of brain [6]. Therefore, selecting optimal electrodes combinations is of great importance for improving classification accuracy of BCI system

In previous study, Kayikcioglu et al. [3] only uses data in electrode A1 of BCI Competition II datasets Ia to analyze and recognize EEG data, and coefficients of the second order polynomial are used for feature extraction. Then, the features are classified using k-nearest neighbor (k-NN) algorithm and obtained 92.15% classification performance. Though classification accuracy shows encouraging result, there is a huge loss of information from other electrodes for EEG signal recognition using only one electrode. Duan et al [7] proposed two feature selection strategies. One is based on combination of optimal single electrodes. This strategy is simple and effective, but classification performance is still weak and optimal electrode is not defined explicitly. The other is based on combination of optimal time spans. This strategy obtained good classification performance, but more tedious. Previously, we also proposed optimal electrodes recombination strategy [8], but did not obtain high classification accuracy.

In this study, we propose a novel EEG signal recognition method. This method uses modified optimal electrodes recombination strategy, and redefines optimal electrodes and optimal electrodes combination explicitly. PCA is used to select the optimal electrodes and EMD is adopted to extract features. Then these features are used as an input to a Naive Bayes classifier, which has two outputs: negative or positive signal. We apply the proposed method to BCI Competition II data sets Ia. The experiment results show that the classification performance of our method outperform the other methods for data set Ia in the BCI competition.

2 Related Works

2.1 Principal Methods

Different motor imagery tasks may activate specific human brain functional areas. Hence, some scalp electrodes are full of effective information and some are not. The

so-called effective information is the frequency or energy imbedded in the signals that changes over time; which is helpful for signal recognition as soon as the task is performed. It is beneficial to do classification from only the effective information and reject inefficient information [9,10]. Redundant information occupies a big part of the raw data, which makes features extraction difficult. Meanwhile the dimension of EEG data is a little bit high. PCA is a common way to analyze high-dimensional data. We consider to use the PCA to select the optimal electrodes and the classification accuracy based on a Naive Bayes classifier is deemed to be the judge standard of optimal electrodes selection.

Empirical mode decomposition(EMD) method is a kind of adaptive signal decomposition method. It can decompose any section of complex time series into a range of different, simple, non-sinusoidal intrinsic mode functions(IMF). The original EEG signal is decomposed by EMD method into a series of IMFs. Each IMF contains the time domain information and instantaneous frequency information of original EEG signal. Because the decomposition method of the EMD method is completely determined by the local frequency information of the original signal, it has strong adaptability and is suitable for processing nonlinear and non-stationary signals. Therefore, it is a good choice to use the EMD method to extract the signal features.

Naive Bayes classifier is a simple probabilistic classifier based on Bayesian theorem (from Bayesian statistics) with strong (naive) independence assumptions between the features [12]. Naive Bayes is not a single algorithm for training such classifiers, but a family of algorithms. Those algorithms base on a common principle: all naive Bayes classifiers assume that the value of a particular feature is independent of the value of any other feature, given the class variable.

2.2 Principal Component Analysis

Principal component analysis (PCA) is a well-established method for feature extraction and dimensionality reduction. Firstly, calculate the covariance matrix of the original sample, then calculate the eigenvalues and eigenvectors of the covariance matrix, and the corresponding eigenvectors are rearranged according to the order of the eigenvalues. Finally, the target dimension is obtained by setting the cumulative contribution rate threshold. The reduced dimension data is obtained by multiplying the original data by the characteristic vector. Therefore, PCA became a common method for removing the eye movement and blink signal interference on the EEG signal. However, PCA cannot guarantee complete deletion of potential noise from the EEG signal which is similar to its own waveform. So PCA is used as an unsupervised method in the process of feature mapping instead of using the information for the classification of internal data. Looking for features extracted by PCA is data effectively represent the direction rather than the data classification.

2.3 Empirical Mode Decomposition

The empirical mode decomposition algorithm is defined during the algorithm process instead of theory formula. The method is essentially a signal smoothing operation. Its

purpose is to decompose the fluctuation or trend of different scales step by step and then produce a series of data sequences with different characteristic scales. Each sequence is called an intrinsic mode function component. Normally, the lowest frequency IMF component is the trend of the original signal or the raw data removes the mean of the data sequence. The test results show that the EMD method is the best way to extract the trend or mean of the data sequence. The basic process of EMD is: firstly, identify all the local extrema of the original data sequence and then using the cubic spline function to fit the original sequence of upper and lower envelope. Using the upper and lower envelope to calculate the mean envelope line; Let the original data sequence subtract the mean envelope data and a new sequence can be obtained. Finally keep modifying the new sequences until it satisfies the two conditions of IMF judgment mentioned above and we can get the data component. The remainder of the new data sequence of the high frequency component is obtained by subtracting the component from the original signal. It is decomposed into second intrinsic mode functions. Repeat the loop until the data sequence cannot be decomposed. Then we get the residual quantity which represents the trend or the mean of the original sequence of the data.

2.4 Naive Bayes classifier

In machine learning, Naive Bayes classifiers are a family of simple probabilistic classifiers based on applying Bayesian theorem with strong (naive) independence assumptions between the features. This algorithm is a classical algorithm, which is widely used in many fields. In this algorithm, the probability distribution function and the overall probability of the sample in different categories is unknown. In order to obtain these two parameters, we always need a large enough sample set but the increased computation will lead to a long training time. In BCI real-time system, EEG signals have small amount of data and request for a short training time which cause many algorithms in BCI unable to play its role effectively in real-time system.

In this article, we firstly let the data dealt with PCA method to select the optimal electrodes and get the combination strategy. Then, using the EMD to modified the complicated EEG data into a simple and Non-sinusoidal IMF. Finally, the Naïve Bayes classifier charges the calculate work and provides the classification accuracy.

3 Framework of Proposed Method

In this paper, a novel EEG signal recognition method using modified optimal electrodes recombination strategy is proposed, and involves five processes: (1) input multi-electrode EEG data, (2) utilize the optimal electrodes selection method using PCA and obtain optimal electrodes, (3) adopt the modified optimal electrodes recombination strategy and obtain optimal electrode combinations, (4) extract features by using EMD, (5) use Naive Bayes classifier and obtain classification results. The framework of the proposed is shown as Fig. 1.

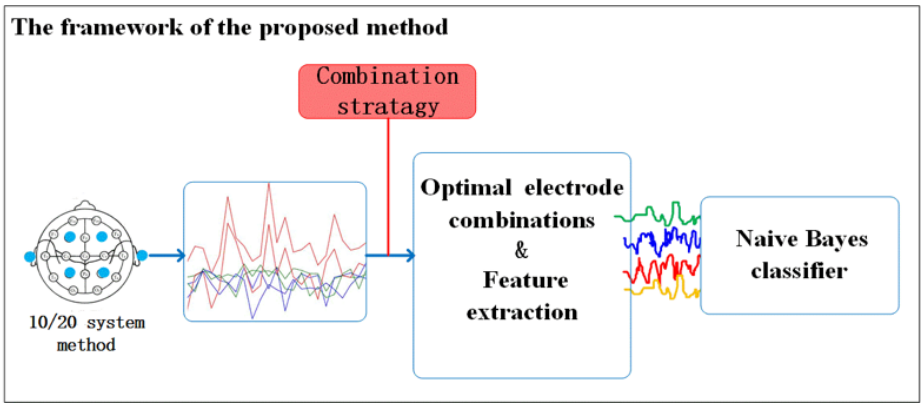

Fig. 1. The framework of the proposed method

Firstly, it is shown that different stimulus will activate special brain function areas. That is to say, only part electrodes of the EEG acquisition device (multi-electrodes cap) are related to the task. Therefore, it is necessary to confirm that which electrodes record the real EEG signals evoked by stimulus. And we called these electrodes as optimal ones. A significant method PCA is used to solve the selection problem.

Secondly, start with an idea that there are neural connections between different brain areas; we discuss the influence on the classification performance by analyzing the multi electrodes, simultaneously based on the optimal electrodes. It can transform the time-frequency information of EEG signals of multi electrodes to the same feature space by joining the properties of multiple electrodes and then carried out the feature extraction algorithm. In the final, it can speculate that whether there is relation between electrodes according to the final classification accuracy.

Thirdly, EEG signal is strong randomness, non-stationary, non-linear and high dimension, and it has high time resolution and low space resolution. Through specific arithmetic operation, EMD method decomposes the signal into a series of Intrinsic Mode Function (IMF) by identifying the oscillation mode contained by the signal itself [10]. Each IMF describes the characteristic of the original signal in time-frequency domain. Moreover, the decomposition of the EMD method completely decided by the local frequency information of the signal. The whole process has strong adaptability and it is very suitable for processing non-stationary and non-linear signals.

At last, the Naive Bayes classifier is used in the paper for classification.

3.1 The optimal electrodes selection method using PCA

The PCA method successively reduce the dimension of the electrodes. Then calculate the cumulative contribution to find out the appropriate parameter. Using the threshold to choose the optimal electrodes. The details of the algorithm are shown in **Algorithm 3-1**.

Algorithm 3-1. The optimal electrodes selection method using PCA

Input: Raw EEG data, Threshold t1 = (1/c + 0.1), where c is the number of label categories in training set.

Output: Optimal electrodes

1. **For** signal E **in** all EEG electrodes:
   E is processed by PCA and calculate the cumulative contribution rate as:

$$r = \frac{\sum_{i=1}^{d} \rho_i}{\sum_{i=1}^{n} \rho_i}$$

   Using the PCA to reduce the dimension of single original EEG signal. The dimension is determined by the cumulative contribution rate of the principal component.
   While d increases:
   if (r >= 85% && r <= 95%)
   the corresponding dimension d is picked up;
   else:
   continue;
2. Using the Bayesian classifier to classify signals in different dimensions;
3. **If** Ac > t1:
   E is optimal electrode.
   Else:
   E is non-optimal electrodes.
4. According to the definition of the advantage electrode, the electrodes are divided into two parts, the dominant electrodes and the non-dominant electrodes.

End

3.2 The modified optimal electrodes recombination strategy

Considering the interaction effect between brain regions, we deal with the multi optimal electrodes simultaneously. Usually, method of exhaustion is used to analyze all the possible combinations. However, it is a big workload and it is not suitable for applications of BCI. According to the physiology research conclusion, different motor imagery tasks have their intrinsic brain areas. Thus the modified optimal electrodes recombination strategy is based on the conclusion of intrinsic brain areas, including four steps as follows.

Algorithm 3-2. The modified optimal electrodes recombination strategy

Input: Raw EEG data, Threshold t2=80%

Output: Optimal combinations and non-optimal combinations

1. Select Top2 electrodes as C_o; Others as C_s;
   C_g = Combination C_o with $\{C \in C_s\}$
2. Using the Bayesian classifier to classify C_g;

Aa = Bayesian classifier (C_g);

3. **If** Aa > t2:

the combination C_g is optimal combination.

Else:

the combination C_g is non-optimal combination.

4. Obtain all the optimal combinations.

End

3.3 Feature extraction using EMD

Any complex time series is composed of several intrinsic mode functions (IMFs), which are different, simple and non-sinusoidal components. So a finite set of IMFs can be separated from the complex time series. Each IMF should satisfy two conditions [11]:

(1) in the whole dataset, the number of extrema and the number of zero-crossings must either be equal or differ at most by one;

(2) at any point, the mean value of the envelope defined by the local maxima and the envelope defined by the local minima is zero.

The process of generating these IMFs is referred to as the EMD decomposition. For the given EEG signal, the detail process of IMF extraction is shown in **Algorithm 3-3**.

Algorithm 3-3. Feature extraction using EMD

Input: Electrode combinations

Output: Features

1. Using $x'(t) = 0$ to identify all the local extrema of $x(t)$. Finding out all of the local maxima point with the cubic spline curve and then connecting them to generate the upper envelope. Lower envelope is generated in the same way. Then, calculate the mean envelope $m_1(t)$.

2. The component $h_1(t)$ is produced by subtracting the $m_1(t)$ from the original data.

3. **If** $h_1(t)$ not satisfies IMF (1)(2) conditions:

Process $h_1(t)$ by step 1,2

Else:

Get the IMF $c_1(t) = h_k(t)$

4. Calculate $r_1(t) = x(t) - c_1(t)$.

5. Let the $r_1(t)$ as a new signal to process (1) - (4) steps.

End

3.4 Classification using Naive Bayes classifier

After extracting the features by EMD we can use the Naive Bayes classifier to classify the EEG data. The input should include the training data, training label, testing data and testing label. The algorithm details are shown as the **Algorithm 3-4**.

Algorithm 3-4. Classification using Naive Bayes classifier

Input: Sample set $X = \{x_1, x_2, \dots, x_N\}$;
The extracted EMD features for x_i are $a_i \in \mathbb{R}^n, (i = 1, 2, \dots, N)$;
Training set labels $T = \{T_1, T_2, \dots, T_c\}$;
Output: Prediction on testing set $L = \{L_1, L_2, \dots, L_u\}$;

1. Compute prior probabilities $P(Y = T_k) = \frac{N_k}{N}$, where N_k is the number of samples in class k;

And compute condition probabilities $P(A = a_{ij} | Y = T_k), (j = 1,2, \dots, n)$ where a_{ij} is the j-th component of feature a_i;

2. Naive Bayes classifier judge X

For m **in** 1:c

Compute

$$P(L_{test}|T_k) = P(T_k) \prod_{j=1}^{n} P\left(a_j^{test}|T_k\right)$$

Output label of naive Bayes classifier is:

$$T_{test} = argmax_{T_k} P(T_k) \prod_{j=1}^{n} P(a_j^{test}|T_k)$$

End

4 EXPERIMENTAL RESULTS

4.1 Dataset description

In this study, we used the publicly available BCI Competition II data sets Ia [13]. The datasets are acquired from a healthy subject. The task of the subject is to move a cursor up and down on a computer screen by imagination. Simultaneously, the cortical potentials of the subject are acquired. When the cortical potentials of the subject are recording, the subject receives visual feedback of their slow cortical potentials. Central parietal region electrode (Cz-Mastoids) is regarded as the reference electrode, and also records their slow cortical potentials. If the slow cortical potentials of the subject are positive, the cursor on the screen is moved up. And if the slow cortical potentials of the subject are negative, the cursor is moved down. EEG signals are collected with 6-channel electrodes. They are, respectively, A1-Cz (A1=left mastoid), A2-Cz, two

cm frontal of C3 (F3), two cm parietal of C3 (P3), two cm frontal of C4 (F4), and two cm parietal of C4 (P4). The sampling rate is 256 Hz. The experiment included 268 trials and each trial lasted six seconds without inter-trial intervals. In addition, there are 561 samples for EEG signal analysis. From 0.5 second to 6 seconds, there is a highlighted goal appeared at the top or the bottom of the screen to indicate negativity or positivity. From 2 second to 5.5 seconds, EEG signals are recorded for training and testing.

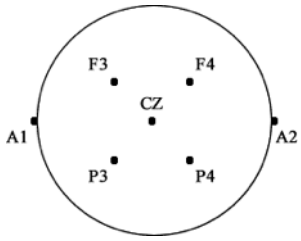

Fig. 2. The distribution diagram of the six electrodes in scalp potentials figure.

4.2 Parameter selection of optimal electrodes selection using PCA

According to step 1 of the optimal electrodes selection method using PCA, 28 different dimensions are obtained, ranging from dimension 3 to dimension 30. From the 28 dimensions, 7 different dimensions is equidistantly selected, namely dimension 3, 5, 10, 15, 20, 25, 30. The raw EEG data of each electrode is reduced to those 7 different dimensions and we obtain 7 EEG data in each electrode. The 7 EEG data of each electrode are classified by using Naive Bayes classifier and we get the 7 accuracies in each electrode, shown as Fig. 2. The average accuracy of 7 accuracies of each electrode is calculated, seen as Table 1. There are two classes in EEG data used, so threshold t1 is set to 60%. The electrode is selected as optimal electrode if its average accuracy is above 60%. In this way, A1, A2, F3 and P3 are optimal electrodes.

Table 1. The average accuracy of each electrode

Electrode	**A1**	**A2**	F3	F4	P3	P4
Average Accuracy (%)	78.89	82.10	65.43	53.63	65.24	54.12

4.3 Parameter selection of the modified strategy

All the optimal electrodes are sorted by the average accuracy and we select the top two electrodes A1 and A2 as intrinsic brain areas electrodes. Then four different combinations that contain intrinsic brain areas electrodes are obtained, which are A1A2, A1A2P3, A1A2F3, and A1A2P3F3, respectively. Similarly, we utilize PCA to reduce the dimension of each combination and Naive Bayes classifier to obtain accu-

racies. The accuracies of each combination in different dimensions are plotted in Fig. 3.

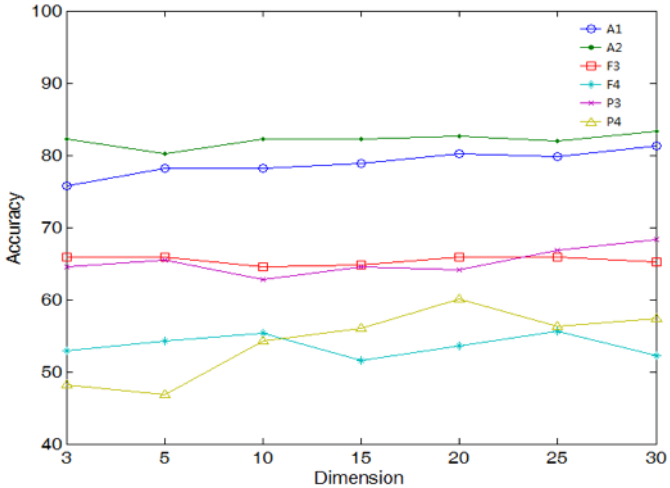

Fig. 3. The classification accuracies of each electrode in 7 different dimensions.

Then the average accuracies of four electrode combinations are obtained, shown in Table 2. The electrode combination is selected as optimal combination if its average accuracy is above 80%. In this way, A1A2, A1A2P3, A1A2F3, and A1A2P3F3 are optimal combinations.

Table 2. The average accuracies of four electrode combinations

Electrode Combination	**A1A2**	**A1A2P3**	**A1A2F3**	**A1A2P3F3**
Average Accuracy (%)	84.64	85.76	87.81	84.69

4.4 Result of the proposed method

For each optimal combination, the corresponding EEG data is extracted features by using EMD method. After that, the data that extracted features are classified by using Naive Bayes classifier. In this way, we get the final accuracies of all the optimal combinations, shown as Table 3.

Table 3. The final classification results of all the optimal combinations

Optimal combination	Classification accuracy(%)
A1A2	85.32
A1A2F3	90.44

A1A2P3	**93.86**
A1A2F3P3	87.03

There is obvious improvement of the classification performance of combination A1A2F3 and A1A2P3. So it is speculated that interaction effect exists between central region and frontal area, while also exists between central region and parietal region. Obviously, the front one is thinner than the last one. The classification performance of combination A1A2F3P3 is relatively weaker. So there is no significant interaction effect between central region, parietal and frontal area. A1 and A2 represent the information of the central electrode, so the performance of the combination A1A2 is close to the performance of the single electrode.

The experimental results obtained by this method are compared with the results of other researchers, and the results are compared with the results shown in the Table 4.

Table 4. Comparison with the results of other researchers

Methods	Classification accuracy(%)
Gamma band power spectrum and SCP + linear classifier [14]	88.70
Alpha band power spectrum and SCP + neural network [15]	91.47
Spectral centroid feature + PCA dimension reduction and Bayesian [16]	90.44
Wavelet transform + probabilistic neural network [17]	90.80
Polynomial fitting + KNN [3]	92.15
LDA-after-PCA+V-ELM [18]	93.52
Advantage combination + Bayesian	**93.86**

The first five lines shows the BCI 2003 Data set Ia data set of previous studies, where the first row shows the first prize results of competition, the EEG data in channel 4 is used for the extraction of gamma band power spectrum information and EEG data in channel 1 is used for the construction of features. The signal is classified by linear classifier method with recognition rate at 88.7%. On the second line, EEG data in channel 4 and channel 6 are used for extraction of α band spectrum information. EEG data in channel 1 is used for construction of features. The signal is classified by neural network method with recognition rate at 91.47%. The third line showed that spectral centroid feature of EEG data in total six channels is extracted to be dimensionally reduced firstly and then classified by Bayesian method with recognition rate at 90.44%. The fourth line illustrates results of recognition with probabilistic neural network method classifying signal after features extraction with wavelet transformation method. On the fifth line, data is analyzed with a combination method of Polynomial fitting and K-nearest neighbor. The algorithm structure is too complicated though the classification performance is improved.

5 Discussion and Conclusion

In this paper, we proposed a novel EEG signal recognition method using modified optimal electrodes recombination strategy. The main conclusions are summarized as follows:

1) The method is applied to the data set Ia of BCI competition, which is also analyzed by other researchers. The winner acquired 88.7% using gamma band power combined with SCPs [14]. The best result (93.52%) is obtained by Duan et al. [15] based on LDA-after-PCA feature extraction method and V-ELM classifier. Our method makes a growth of 0.34% than the best one, which is simple and efficient.

2) It is significant to select optimal electrodes. The results show that optimal electrodes selected by the proposed method are consistent with electrodes selected by psychology knowledge, furthermore, it is subtle.

In a word, the proposed method offers great potential for improving the classification accuracy of BCI datasets. What is more, there are some shortcomings should be overcome in the future. The data set Ia of BCI competition includes 6 electrodes only. If much more electrodes are included, the amount of calculation and complexity will increase. The concurrent computation and high-performance equipment may be expected to be an assistant.

The idea of this method is based on the brain processing towards tasks, considering the phenomenon that multiple regions of brain could be activated at the same time. Therefore, the relationship of different brain regions is deduced from analyzing the relationship of EEG data collected on corresponding regions by recording electrodes. This method combines the knowledge of psychology and information technology and it provided a new way of studying interactions between stimulating brain regions with certain significance.

Acknowledgements. This research is partially sponsored by Natural Science Foundation of China (Nos. 61672070, 81471770, 61572004), the Beijing Municipal Natural Science Foundation (grant number 4182005).

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